

Homework: Problem 1

Farmer problem

- Consider a European farmer who specializes in raising 1) wheat, 2) corn, and 3) sugar beets on his 500 acres.
- During the winter, he wants to decide how much land to devote to each crop.
- At least 200 tons (T) of wheat and 240 T of corn are needed for cattle feed.
- These amounts can be raised on the farm or bought from a wholesaler.
- Any production in excess of the feeding requirement would be sold.
- Over the last decade, the mean selling prices are shown in Table below. The purchase prices are 40% more than this due to the wholesaler's margin and transportation costs.
- Another profitable crop is sugar beet, which he expects to sell at \$36/T; however, the European Commission imposes a quota. Any amount in excess of the quota can be sold only at \$10/T. The farmer's quota for next year is 6000 T.
- Based on past experience, the farmer knows that the mean yield on his land is roughly 2.5 T, 3 T, and 20 T per acre for wheat, corn, and sugar beets, respectively.

	Wheat	Corn	Sugar Beets
Yield (T/acre)	2.5	3	20
Planting cost (\$/acre)	150	230	260
Selling price (\$/T)	170	150	36 under 6000 T 10 above 6000 T
Purchase price (\$/T)	238	210	—
Minimum requirement (T)	200	240	—
Total available land: 500 acres			

x_j = acres of land devoted to crop j , $j = 1, 2, 3$

w_1 = tons of wheat sold

y_1 = tons of wheat purchased

w_2 = tons of corn sold

y_2 = tons of corn purchased,

w_3 = tons of sugar beets sold at the favorable price

w_4 = tons of sugar beets sold at the lower price

$$\begin{aligned} \min z = & 150x_1 + 230x_2 + 260x_3 && \text{(planting cost)} \\ & + 238y_1 + 210y_2 && \text{(purchase cost)} \\ & - 170w_1 - 150w_2 - 36w_3 - 10w_4 && \text{(revenue)} \end{aligned}$$

subject to:

$$x_1 + x_2 + x_3 \leq 500 \quad \text{(land avail)}$$

$$2.5x_1 + y_1 - w_1 \geq 200 \quad \text{(wheat req)}$$

$$3x_2 + y_2 - w_2 \geq 240 \quad \text{(corn req)}$$

$$w_3 + w_4 \leq 20x_3 \quad \text{(beets sold)}$$

$$w_3 \leq 6000 \quad \text{(sugar beet quota)}$$

$$x_1, x_2, x_3, y_1, y_2, w_1, w_2, w_3, w_4 \geq 0$$

Throughout the formulation, maintain the decision variable order

$$x = [x_1, x_2, x_3, y_1, y_2, w_1, w_2, w_3, w_4]^T$$

Use matrix notation to write the LP

$$\min \{c^T x : A x \geq b\} \quad (*)$$

- Convert all constraints into $\geq$ form to match the structure in (*).
- Specify c , A , b .
- Solve the LP model.

Homework: Problem 2

Fractional knapsack problem

- The cash required for each investment and the net present value (NPV) each investment adds to the firm are listed in the table below.
 - The return on investment (ROI in %) for investment 1 is

$$\frac{5.6-5}{5} \times 100 = \left(\frac{5.6}{5} - 1\right) \times 100 = (1.12 - 1) \times 100 = 0.12 \times 100 = 12\%$$
- Partial investments are allowed. (In week 04, partial investment will NOT be allowed.)
- The cash available for investment is \$15,000 = \$15 K.

j	w_j (\$ K)	c_j (\$K)	
Investment	Cash Required	NPV	ROI
1	\$5.0	\$5.6	12.0%
2	\$2.4	\$2.7	12.5%
3	\$3.5	\$3.9	11.4%
4	\$5.9	\$6.8	15.3%
5	\$6.9	\$7.7	11.6%
6	\$4.5	\$5.1	13.3%
7	\$3.0	\$3.3	10.0%
SUM	\$31.2		

Decision variables:

x_j = fraction of investment j chosen

$$\text{Maximize } z = \sum_{j=1}^7 c_j x_j$$

subject to:

$$\sum_{j=1}^7 w_j x_j \leq 15$$

$$x_j \in \{0,1\} \text{ for } j = 1, 2, \dots, 7$$